

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: TBT3 Support on USB4 Dock UFP Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Removes the mandatory requirement to support TBT3 on a USB4 Dock UFP.
Removes the *TBT3 Not Supported* bit

Benefits as a result of the changes:

When the USB4 Version 1 was written, in 2018 timeframe, it was decided to ensure interoperability with the eco system at that time, which was only TBT3. The decision was taken with the intention of revisiting this after 5 years.

While the support for TBT3 Devices is expected to be preserved for many years, the support for TBT3 Host can be relaxed.

Now, ~5 years later, the ask is to ease the complexity of a USB4 Dock by removing the requirement to support TBT3 on the UFP. By doing so, the USB4 Dock:

- Does not need to support the special Configuration Spaces as it will not work with a TBT3 CM
- Does not need to implement an xHC

The *TBT3 Not Supported* bit is not usable, as when it needs to be read there is no USB4 Link.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

None

An analysis of the hardware implications:

None

An analysis of the software implications:

None

An analysis of the compliance testing implications:

Update the CTS accordingly

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Actual Change

(a). 13 Interoperability with Thunderbolt™ 3 (TBT3) Systems

13 Interoperability with Thunderbolt™ 3 (TBT3) Systems

This chapter defines requirements for a USB4 Product to be TBT3-Compatible. A USB4 Product that is TBT3-Compatible can operate in a USB4® Fabric that includes any combination of the following:

- A Thunderbolt 3 Router.
- A Thunderbolt 3 Connection Manager.
- A Thunderbolt 3 Active Cable.

A USB4 Host and USB4 Peripheral Device may optionally support TBT3-Compatibility. If a USB4 Host or USB4 Peripheral Device supports TBT3-Compatibility, it shall do so as defined in this chapter.

A USB4 Hub shall support TBT3-Compatibility as defined in this chapter. A USB4 Hub shall support TBT3-Compatibility on all of its Downstream Facing Ports. ~~If the USB4 Hub is a USB4-Based Dock, it shall support TBT3-Compatibility on its Upstream Facing Port in addition to all its Downstream Facing Ports.~~ A USB4 Hub may optionally support TBT3-Compatibility on its Upstream Facing Port.

~~A Router uses the TBT3 Not Supported bit to indicate whether it is TBT3-Compatible. This chapter does not apply to a Router with the TBT3 Not Supported bit set to 1b.~~

(b). Table 8-3 Router Configuration Space Basic Attributes

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
6	ROUTER_CS_6	1	TBT3 Not Supported (TNS) <u>Reserved</u> A Router shall set this field to 1b if it does not support the TBT3-Compatible behavior defined in Chapter 13. A Router shall set this field to 0b if it supports the TBT3-Compatible behavior defined in Chapter 13. Deprecated filed (TBT3 Not Supported)	RORs <u>vd</u>	Vendor Defined

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(c). 13.4 Configuration Layer

This section only applies to [a Device Router that supports TBT3-Compatibility on its Upstream Facing Port](#).

(d). 13.4.4 Notification packet

~~This section only applies to a Device Router that supports TBT3-Compatibility on its Upstream Facing Port.~~

(e). 13.4.5 Bit Banging Interface

~~This section only applies to a Device Router that supports TBT3-Compatibility on its Upstream Facing Port.~~

(f). 2.1.5 Thunderbolt™ 3 (TBT3) Compatibility Support

A USB4 Host or USB4 Peripheral Device can optionally support interoperability with Thunderbolt 3 (TBT3) products.

A USB4 Hub is required to support interoperability with Thunderbolt 3 products on all of its Downstream Facing Ports. ~~A USB4 Based Dock is required to support interoperability with Thunderbolt 3 products on its Upstream Facing Port in addition to all of its Downstream Facing Ports.~~ [A USB4 Hub may optionally support TBT3-Compatibility on its Upstream Facing Port.](#)

When interoperating with a TBT3 product, Thunderbolt Alt Mode is established on the link between products. The USB Type-C Specification describes how a USB4 product negotiates and enters Thunderbolt Alt Mode.

Chapter 13 describes the additional requirements for a Router that supports TBT3-compatible interoperability.